

## MAPLE CREEK, SK, Canada

WMO: 714530

Lat: 49.9025N

Lon: 109.4661W

Elev: 767

StdP: 92.45

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -30.4      | -26.7 | -33.8                         | 0.2 | -30.3 | -29.9 | 0.3 | -26.7 | 13.9                  | -1.3 | 12.2 | 0.7  | 1.7                   | 80   | 0.673 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.8                   | 32.3            | 17.3 | 30.4 | 17.0 | 28.4 | 16.5 | 19.4                | 27.6 | 18.4 | 26.7 | 17.4 | 25.8 | 4.7                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.8                            | 13.2 | 22.2 | 15.5 | 12.1 | 21.2 | 14.3 | 11.2 | 20.1 | 58.8          | 28.0 | 55.0 | 26.5 | 51.9 | 25.7 | 24.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.5          | 10.0 | 8.8 | -35.7                      | 35.0 | 3.3                | 1.8 | -38.0                                              | 36.2 | -39.9      | 37.3 | -41.8      | 38.3 | -44.1      | 39.5 |    |    |
| (5)               |      |     | -35.8                      | 21.3 | 3.2                | 1.6 | -38.1                                              | 22.4 | -40.0      | 23.4 | -41.8      | 24.3 | -44.2      | 25.5 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 5.4   | -8.3  | -7.0  | -1.0  | 5.5   | 10.9  | 15.3  | 19.4  | 18.4  | 13.2  | 5.9   | -1.4  | -7.2 | (6)  |
| (7)  |                                                                                  | DBStd       | 11.69 | 9.44  | 8.53  | 8.01  | 5.18  | 4.61  | 3.55  | 3.45  | 3.80  | 4.88  | 5.42  | 7.49  | 8.76 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2631  | 568   | 477   | 342   | 150   | 44    | 3     | 0     | 1     | 22    | 147   | 344   | 533  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4867  | 826   | 710   | 599   | 385   | 235   | 104   | 27    | 46    | 167   | 385   | 592   | 791  | (9)  |
| (10) |                                                                                  | CDD10.0     | 943   | 0     | 0     | 2     | 15    | 72    | 161   | 293   | 262   | 117   | 21    | 2     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 139   | 0     | 0     | 0     | 0     | 4     | 12    | 62    | 49    | 12    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 2234  | 0     | 0     | 0     | 6     | 77    | 209   | 916   | 788   | 231   | 9     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 806   | 0     | 0     | 0     | 1     | 13    | 53    | 355   | 307   | 76    | 1     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.9   | 3.7   | 3.6   | 4.2   | 4.4   | 4.2   | 3.8   | 3.6   | 3.5   | 3.6   | 4.0   | 3.9   | 3.8  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 437   | 23    | 18    | 26    | 35    | 48    | 88    | 50    | 45    | 37    | 28    | 21    | 20   | (15) |
| (16) |                                                                                  | PrecMax     | 642   | 61    | 34    | 49    | 79    | 114   | 225   | 98    | 123   | 81    | 96    | 48    | 38   | (16) |
| (17) |                                                                                  | PrecMin     | 219   | 4     | 3     | 3     | 12    | 3     | 32    | 8     | 0     | 0     | 3     | 5     | 1    | (17) |
| (18) |                                                                                  | PrecStd     | 89    | 14    | 10    | 12    | 18    | 29    | 45    | 26    | 32    | 23    | 22    | 12    | 9    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 11.8  | 12.5  | 18.3  | 23.6  | 28.5  | 31.3  | 34.4  | 34.5  | 32.0  | 24.7  | 18.1  | 12.3 | (19) |
| (20) |                                                                                  |             | MCWB  | 6.3   | 5.8   | 8.9   | 11.9  | 15.1  | 17.3  | 18.1  | 17.5  | 16.0  | 13.3  | 9.1   | 5.5  | (20) |
| (21) |                                                                                  | 2%          | DB    | 8.8   | 9.0   | 15.0  | 20.6  | 25.5  | 28.0  | 32.5  | 32.2  | 28.9  | 20.9  | 14.5  | 9.4  | (21) |
| (22) |                                                                                  |             | MCWB  | 4.1   | 4.2   | 7.4   | 10.7  | 14.1  | 17.1  | 18.0  | 16.6  | 14.9  | 11.4  | 7.6   | 4.2  | (22) |
| (23) |                                                                                  | 5%          | DB    | 6.6   | 6.5   | 12.2  | 17.8  | 23.0  | 25.6  | 30.6  | 30.3  | 26.0  | 17.9  | 11.8  | 7.0  | (23) |
| (24) |                                                                                  |             | MCWB  | 3.2   | 2.7   | 5.9   | 9.0   | 12.9  | 16.0  | 17.9  | 16.5  | 14.2  | 9.9   | 6.0   | 2.9  | (24) |
| (25) |                                                                                  | 10%         | DB    | 4.7   | 4.5   | 9.6   | 15.2  | 20.4  | 23.5  | 28.4  | 28.0  | 23.0  | 15.1  | 9.0   | 4.7  | (25) |
| (26) |                                                                                  |             | MCWB  | 1.8   | 1.3   | 4.3   | 7.6   | 11.6  | 14.9  | 17.1  | 16.1  | 12.9  | 8.4   | 4.3   | 1.6  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 6.9   | 6.3   | 9.8   | 12.8  | 16.6  | 20.2  | 22.1  | 19.7  | 17.6  | 13.9  | 10.1  | 6.7  | (27) |
| (28) |                                                                                  |             | MCDWB | 11.0  | 11.7  | 16.7  | 21.7  | 24.6  | 27.2  | 30.0  | 27.9  | 28.1  | 23.0  | 17.7  | 11.5 | (28) |
| (29) |                                                                                  | 2%          | WB    | 4.7   | 4.4   | 7.8   | 11.1  | 14.8  | 18.2  | 20.1  | 18.6  | 16.0  | 11.9  | 8.0   | 4.6  | (29) |
| (30) |                                                                                  |             | MCDWB | 8.2   | 8.5   | 14.2  | 19.4  | 23.3  | 25.0  | 28.6  | 27.2  | 26.2  | 19.6  | 14.0  | 8.6  | (30) |
| (31) |                                                                                  | 5%          | WB    | 3.2   | 2.9   | 6.2   | 9.6   | 13.6  | 16.9  | 18.9  | 17.7  | 14.7  | 10.3  | 6.2   | 3.0  | (31) |
| (32) |                                                                                  |             | MCDWB | 6.3   | 6.3   | 11.7  | 16.8  | 21.8  | 23.9  | 27.3  | 26.7  | 24.1  | 16.8  | 11.4  | 6.7  | (32) |
| (33) |                                                                                  | 10%         | WB    | 1.9   | 1.4   | 4.6   | 8.1   | 12.2  | 15.6  | 17.9  | 16.8  | 13.5  | 8.9   | 4.5   | 1.5  | (33) |
| (34) |                                                                                  |             | MCDWB | 4.8   | 4.3   | 9.2   | 14.2  | 19.1  | 22.1  | 26.2  | 25.8  | 21.8  | 14.3  | 8.6   | 4.7  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 12.1  | 11.9  | 11.7  | 13.1  | 13.3  | 12.6  | 14.8  | 14.9  | 14.1  | 12.8  | 11.7  | 11.6 | (35) |
| (36) |                                                                                  |             | MCDBR | 12.3  | 13.4  | 15.3  | 17.3  | 17.4  | 16.5  | 17.9  | 18.9  | 18.9  | 16.8  | 14.8  | 12.2 | (36) |
| (37) |                                                                                  |             | MCWBR | 8.8   | 9.3   | 8.9   | 9.3   | 8.4   | 7.8   | 7.2   | 7.3   | 8.2   | 9.1   | 9.0   | 8.3  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 12.3  | 13.3  | 14.7  | 16.4  | 15.5  | 14.9  | 15.8  | 15.6  | 17.1  | 15.7  | 14.4  | 12.1 | (38) |
| (39) |                                                                                  |             | MCWBR | 9.1   | 9.6   | 9.1   | 9.2   | 7.9   | 7.7   | 7.3   | 6.9   | 8.2   | 8.8   | 9.2   | 8.7  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.270 | 0.269 | 0.298 | 0.352 | 0.377 | 0.362 | 0.369 | 0.388 | 0.338 | 0.299 | 0.268 | 0.256 | (40) |      |
| (41) |                                                                                  | taud        | 2.371 | 2.441 | 2.438 | 2.299 | 2.272 | 2.379 | 2.370 | 2.288 | 2.404 | 2.496 | 2.477 | 2.392 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 786   | 893   | 918   | 892   | 880   | 895   | 883   | 844   | 858   | 842   | 792   | 754   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 64    | 76    | 92    | 118   | 127   | 115   | 115   | 119   | 95    | 72    | 57    | 54    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.28  | 2.29  | 3.62  | 4.88  | 5.83  | 6.26  | 6.80  | 5.54  | 4.01  | 2.49  | 1.41  | 0.99  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.13  | 0.18  | 0.23  | 0.33  | 0.39  | 0.41  | 0.41  | 0.40  | 0.34  | 0.26  | 0.12  | 0.09  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (29 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page